

RICHARD PHONG

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EXPERIENCE

EOG Resources – Software Developer

2025 - Present

- Developed a React/Redux web application serving as a proxy model for reservoir simulation, reducing per-run time from **40-50s** to **1-2s** enabling a projected **5% increase** in output across **1000+** future wells
- Engineered reusable visualization components, including **time-series charts, key metrics dashboards, and geospatial well maps** using SciChart and MapBox to surface **100,000+** data points.
- Established an AI-forward development workflow by implementing strict **ESLint** rulesets and **Datadog** Synthetic tests within a **GitHub Actions CI/CD** pipeline, ensuring automated quality gates for AI-generated code
- Led **cross-team integration** of gun barrel visualization, coordinating library versioning across codebases and debugging partner team code to deliver on schedule despite limited resources

EOG Resources – Software Developer Intern

2024

- Created an anomaly detection pipeline, using both rule-based and ML-based methods on device logs.
- Leveraged **Elastic** along with internal ticketing and security APIs to escalate anomalies in real time.

NASA – Software Developer Intern

2023 - 2024

- Redesigned and refactored the task categorization and scheduling interface for an ISS crew calendar application used by astronauts during Artemis III preparation, improving page responsiveness and navigation flow.
- Built an API to automate On-Board Training feedback delivery, eliminating a manual email workflow.

Paycom – Software Developer Intern

2023

- Created an ASP.NET/React WebAPI using FedEx APIs to create labels, streamlining package processing for over **15,000** packages weekly.
- Proactively established a unit testing suite using XUnit, MOQ, and Jest, achieving 90% code coverage across backend and frontend to ensure reliability ahead of production rollout.

NASA – Software Developer Intern

2022 - 2023

- Developed a Razor Pages tool enabling real-time task coordination between ground crew and ISS crew members.
- Refactored code for an ASP.NET-based program used to train astronauts boarding the ISS, reducing file sizes by upwards of **60%** and improving script speed by over **5x**.

PROJECTS

Game Engine | github.com/rphong/tengin

2023

- Created a game engine in C++ using OpenGL to recreate Wii Tanks! as a personal project.
- Implemented a collision system using SAT with an automated hitbox creation using a convex hull algorithm.

HackTX 2023: Savorr | devpost.com/software/savorr

2023

- Created a web application to save on groceries using Next.js/Tailwind/Vercel as a hackathon project.
- Retrieved and displayed lowest prices from Kroger API and suggested ingredient substitutions using OpenAI API.
- Awarded Best Use of Github amongst 700+ students and nearly 200 submissions.

Froggie Adventures | github.com/rphong/Froggie

2022

- Created a 2-D procedurally generated platformer with Unity and C#, leading a team of 3 as a club project.
- Developed a procedural generation algorithm to dynamically create levels that scale in difficulty.
- Displayed to 200+ interested students at the annual Cat's Back (UH club fair).

EDUCATION

University of Houston – Bachelor of Science in Computer Science

2024

SKILLS

Languages: Typescript, C++, C#, Java, HTML, CSS, Python

Tools: Unity, Git/GitHub, Github Actions, Blender, Figma, Elastic, Datadog

Frameworks/Libraries: Redux, Plotly, MapBox, SciChart, Express, React, .NET, Node, Maui, Xamarin, OpenGL